



A high-precision and efficient method for badminton action detection in sports using You Only Look Once with Hourglass Network

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ABSTRACT

Standardized striking movements are essential in badminton for enhancing player techniques and minimizing sports-related injuries. However, accurately detecting these movements against complex backgrounds while balancing precision and speed remains a significant challenge. To address this, we propose a novel model that synergizes You Only Look Once (YOLO) with the Hourglass Network (HGNet), called YOLO-HGNet, to enhance feature learning across multiple levels. By replacing traditional convolutional modules with Depth-Wise Convolution (DWConv), we achieve significant improvements in data processing efficiency. Additionally, our model incorporates a combination of self-attention and convolution mechanisms (ACmix) and FocalModulation to improve object localization and recognition accuracy in complex backgrounds. Our method leverages action vectors and machine learning techniques to accurately detect and classify six key badminton strokes: backhand push, backhand net shot, forehand clear, forehand push, forehand lift, and forehand net shot. Empirical evaluations demonstrate that our approach achieves a mean Average Precision (mAP) of 96.1% for detecting badminton player postures, outperforming existing advanced methods by at least 8.8%. Furthermore, our method achieves an average accuracy of 95.4% in classifying the six badminton strokes. These results underscore the superior capability of YOLO-HGNet for precise and efficient pose detection, recognition, and classification of badminton strokes, contributing significantly to advancements in sports science and athlete training methodologies.

1. Introduction

Mastering standardized techniques in badminton is critical for players at all skill levels to enhance agility and overall gameplay. These techniques are vital for intercepting the shuttlecock before it descends below the net height, thereby conferring a strategic advantage during matches (Lin et al., 2024; Vijay Singh Dr. Sameer, 2023). A comprehensive analysis of player postures and movements, derived from badminton footage, categorizes striking actions into four primary types: clears, drop shots, drives, and smashes (Chu and Situmeang, 2017; Ramasinghe et al., 2014). For beginners, mastering these actions is essential for building a solid technical foundation, while advanced players can leverage this understanding to refine their power and

performance on the court. This approach underscores the necessity of detailed analysis in badminton dynamics, which is instrumental in developing effective coaching strategies and enhancing athlete training (He et al., 2024; Liu et al., 2017).

The advent of the digital age has propelled artificial intelligence (AI) to new heights, enabling the processing and interpretation of visual data with accuracy approaching that of human vision. The field of computer vision (CV), particularly in the recognition of human actions, has gained significant traction, impacting various domains including sports analytics (Host and Ivašić-Kos, 2022), security (Zhao et al., 2021, 2023), and educational practices. The emergence of intelligent badminton action recognition technology represents a transformative shift, offering objective analysis that significantly enhances training efficacy

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(Feichtenhofer et al., 2017; Huang et al., 2021; Liu and Ke, 2020).

Despite these technological advancements, current methodologies in badminton action recognition encounter significant limitations. Key challenges include achieving high levels of accuracy, maintaining swift processing speeds, and capturing the intricate nuances of rapid and diverse movements intrinsic to the sport. The complexity involved in modeling such dynamic and varied actions underscore a critical research gap. This gap highlights the necessity for the development of more sophisticated recognition systems that not only improve accuracy and processing efficiency but also capture the detailed subtleties of player movements. Addressing these issues is essential for advancing the precision and effectiveness of action recognition technologies in badminton.

To address these challenges, we propose a novel method for recognizing badminton players' hitting movements using the state-of-the-art YOLOv8 model. The main features and contributions of our method are as follows. Firstly, we introduce HGNet as the backbone network of our model to enhance the detection efficiency and accuracy of the overall system. Secondly, our method improves the model's data processing capability by integrating a new attention mechanism, ACmix, into feature-rich nodes. Thirdly, we introduce a state-of-the-art feature processing method into the model's feature processing node to enhance its feature processing ability. Fourth, we compare our proposed method with other action detection methods, including YOLOv5 and Openpose. Finally, we used a motion vector combined with a machine learning method to classify and recognize six kinds of actions.

The main objectives and contributions of this paper are as follows.

- Introducing HGNet into the backbone network of YOLOv8 to improve the efficiency and accuracy of the model.
- The ACmix module is introduced into the feature-rich stage, which significantly improves the accuracy of the model in recognizing the badminton hitting action. This integration helps to extract key features in more detail, which enriches the analytical depth of the model.
- The original Conv was replaced with DWConv to optimize the model parameters and improve the detection efficiency of the model.
- Utilizing FocalModulation to emphasize key regions in the image further, improving the model's recognition capabilities.
- Addressing the empirical requisites of this research, a bespoke dataset comprising 3000 images annotated with keypoints has been meticulously compiled. This dataset, designed to support 13 human keypoints, is instrumental in enabling precise pose estimation within the context of badminton.
- Employing motion vectors to locate and interpret extracted action information effectively facilitates recognition.

The organization of this paper is methodically structured to facilitate ease of understanding and engagement with our findings. Section 2 offers a concise review of the relevant literature, setting the stage for introducing our methodology, colloquially termed "YOLO-HGNet," in Section 3. Section 4 presents a detailed analysis and evaluation of our technique's performance, underpinned by rigorous empirical evidence. Finally, Section 5 concludes our study, summarizing our contributions and outlining avenues for future research endeavors.

2. Related works

Object detection technology is crucial in computer vision, aiming to identify and locate objects within images precisely. The YOLO (You Only Look Once) algorithm series, a cornerstone in object detection, has gained widespread application due to its superior real-time performance and accuracy.

At its core, the YOLO (Redmon et al., 2016) algorithm employs a singular convolutional neural network to predict multiple bounding boxes and class probabilities concurrently, achieving a delicate balance

between speed and accuracy. YOLOv2 (Redmon and Farhadi, 2017) marks a significant evolution from the original YOLO by adopting the DarkNet-19 network architecture instead of GoogleNet. Advancing further, YOLOv3 (Redmon and Farhadi, 2018) further strengthens the feature extraction ability of the model. It improves the detection effect of small objects by introducing a deeper DarkNet-53 as the backbone network and using multi-scale prediction. YOLOv4 (Bochkovskiy et al., 2020) continues the series' legacy of high-speed detection while enhancing model accuracy and robustness. This enhancement is facilitated through the integration of CSPNet, adaptive anchor boxes, Ciou Loss, and the Mish activation function, among other technological advancements. YOLOv5 (Jocher, 2020) inherits the main features of the YOLO family. It provides multiple versions in terms of model size, inference speed, and accuracy to meet different application requirements and computational resource constraints. The latest in the series, YOLOv8 (Jocher et al., 2023), released in early 2023, builds upon the foundation laid by YOLOv5. It introduces comprehensive improvements in performance, speed, and accuracy, which are discussed in this article.

In the contemporary realm of sports science, action recognition in badminton has seen remarkable advancements, greatly enriching athletes' training and performance optimization. A seminal approach employed dense trajectories and trajectory-aligned Histogram of Oriented Gradients (HOG) features, combined with a Support Vector Machine (SVM) classifier, to categorize RGB-D data into four primary action types: smash, forehand, backhand, and service break (Ramasinghe et al., 2014). Further advancing the field, developing a novel skeleton-based deformation model, termed TemPose, has been instrumental in fine-grained motion recognition. This model significantly augments the granularity of action understanding in badminton, allowing for a more detailed examination of player techniques (Magnus et al., 2023). The SwingNet method adeptly captures the salient features and patterns inherent in badminton striking actions, thereby improving action recognition accuracy (Chen, 2023). Additionally, implementing an automated service fault detection system, which integrates computer vision and machine learning through the application of the YOLOv5 object detection model, exemplifies the practical application of these research findings (Goh et al., 2023). In another noteworthy study, the classification of badminton match actions was enhanced by deploying an improved I3D network, namely the SI3D network (Zheng et al., 2023). This innovative network adaptation further enriches the badminton action recognition research landscape, offering profound insights into player dynamics and performance strategies. Further studies have classified the actions in the field by using random forests (Liu, 2024) and Convolutional neural networks (CNN) (Chen, 2024; Yue et al., 2024).

In the ongoing quest to refine object detection within the dynamic environment of sports analytics, particularly in understanding player movements and strategies, existing models have faced challenges, including suboptimal detection accuracy, computational efficiency, and occasional oversight of critical action details. Our study confronts these challenges head-on by introducing YOLO-HGNet, a streamlined detection model that builds upon the robust foundation of YOLOv8. By incorporating innovative elements such as the ACmix attention mechanism, FocalModulation modules, and advanced modules like DWConv, we aim to enhance the capabilities of YOLOv8 significantly.

3. Materials and methods

3.1. Human posture estimation

YOLO-Pose (Maji et al., 2022) associates all keypoints of an individual's pose with a singular anchor point, streamlining the estimation process and enhancing the model's efficiency and scalability. This novel approach mitigates the computational and complexity burdens typical of traditional methods, thereby offering a more robust solution for real-time applications. Building on the foundational principles of

YOLO-Pose, this research endeavors to refine human pose estimation specifically for the domain of badminton. By identifying and focusing on 13 primary keypoints pivotal to capturing the essence of badminton players' movements, this study aims to advance the accuracy and applicability of action pose estimation within the sport. This targeted selection of keypoints is instrumental in developing a nuanced understanding of player dynamics, facilitating a more precise analysis of badminton actions.

Within video frames, this paper defines 13 crucial human body keypoints and their corresponding spatial coordinates, where l denotes the left side, r denotes the right side, t denotes the upper boundary and b denotes the lower boundary. These keypoints include the head $h_n(x_n^h, y_n^h)$, shoulders $ls_n(x_n^{ls}, y_n^{ls})$ and $rs_n(x_n^{rs}, y_n^{rs})$, elbows $le_n(x_n^{le}, y_n^{le})$ and $re_n(x_n^{re}, y_n^{re})$, wrists $lw_n(x_n^{lw}, y_n^{lw})$ and $rw_n(x_n^{rw}, y_n^{rw})$, hips $lh_n(x_n^{lh}, y_n^{lh})$ and $rh_n(x_n^{rh}, y_n^{rh})$, knees $lk_n(x_n^{lk}, y_n^{lk})$ and $rk_n(x_n^{rk}, y_n^{rk})$, and ankles $la_n(x_n^{la}, y_n^{la})$ and $ra_n(x_n^{ra}, y_n^{ra})$, as illustrated in Fig. 1. The origin is located at the top left corner of the image, with the outer frame representing the size of the video frame image and the inner frame representing the detection box for the player. The four vertex coordinates of the detection box are denoted as $(bbox_l, bbox_t)$, $(bbox_r, bbox_t)$, $(bbox_l, bbox_b)$, $(bbox_r, bbox_b)$.

To elucidate the relational dynamics among these keypoints, the study establishes a hierarchical structure with the head and hips classified as the core segments of the body framework. The segments extending from the shoulder to the elbow and from the elbow to the wrist are defined as the upper arm and forearm regions, respectively. Similarly, the segments from the hips to the knee and from the knee to the ankle are identified as the thigh and calf regions, respectively. This hierarchical segmentation is instrumental in enabling a nuanced and intricate analysis of human posture, particularly within the domain of sports kinetics. It allows for a focused examination of the movement mechanics and dynamics of each body region during the execution of badminton strokes, thereby facilitating a comprehensive understanding of player postures and movements.

3.2. YOLO-HGNet model architecture

YOLO-HGNet stands as a pivotal enhancement to the YOLOv8 model, tailored explicitly for detecting badminton player poses with high efficiency and accuracy. Its architectural design is illustrated in Fig. 2. This model incorporates cutting-edge architectural elements and attention mechanisms to refine its computational efficiency and feature extraction

capabilities, setting a new standard in sports analytics and player performance assessment.

The YOLO-HGNet is primarily composed of four parts: the backbone network, the neck network, the detection head, and predict. The HGNet (Lv et al., 2023) network is used as the backbone network, and Focal-Modulation modules are integrated to enhance attention to the key area of the image, effectively suppress irrelevant information, and significantly improve the accuracy of badminton hitting action recognition. The neck network integrates the ACmix attention mechanism to improve further the model's ability to capture local features.

3.2.1. Optimize data processing

The integration of the HGBlock (Lv et al., 2023) module into the YOLOv8 architecture significantly amplifies its capabilities, especially in addressing the complex requirements of pose detection tasks. This strategic enhancement is designed to overcome the limitations inherent in the original c2f (coarse to fine) module, which struggled with capturing and processing feature variations across a broad spectrum of abstraction levels. The HGBlock, with its sophisticated design, allows for a nuanced approach to feature extraction, enabling the model to discern and utilize information from both primary and complex visual cues within an image. Its structure is shown in Fig. 3. This module enhances the model's ability to detect complex postures by improving the recognition and utilization efficiency of various levels of features within images. As a result, YOLOv8 with the HGBlock exhibits significant improvements in both processing speed and detection accuracy.

3.2.2. Lightweight network optimization

The incorporation of Depth-Wise Convolution (DWConv) (Chollet, 2017) into the YOLOv8 architecture addresses a critical issue that plagued the original model: the significant redundancy in intermediate feature maps produced by traditional convolution operations. This redundancy escalated the computational cost and hampered the model's processing speed and efficiency. By deploying DWConv, as detailed in the structural depiction in Fig. 4, the model sees a notable optimization in handling feature maps, thereby enhancing both performance and efficiency.

DWConv diverges from the standard convolution technique primarily in the application of its kernels. Traditional convolutions amalgamate information across all channels of the input feature map, with each kernel generating a new feature map channel through this comprehensive convolution operation. While this process is effective for certain

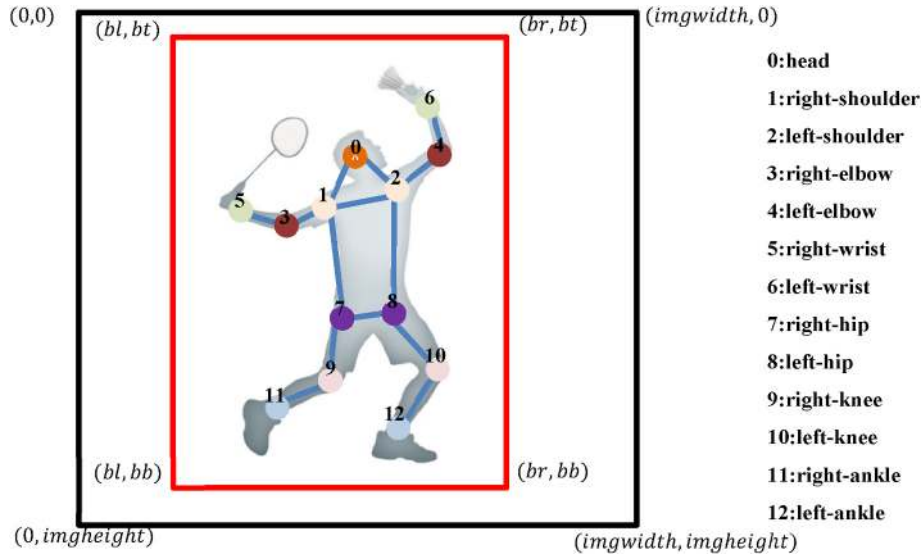


Fig. 1. Major joints. After analyzing a large number of badminton videos, it is found that the other joint points of the head are not the main ones, so 13 main joint points are selected for action detection, and other detailed joints of the head, including eyes, ears, and nose, are removed.

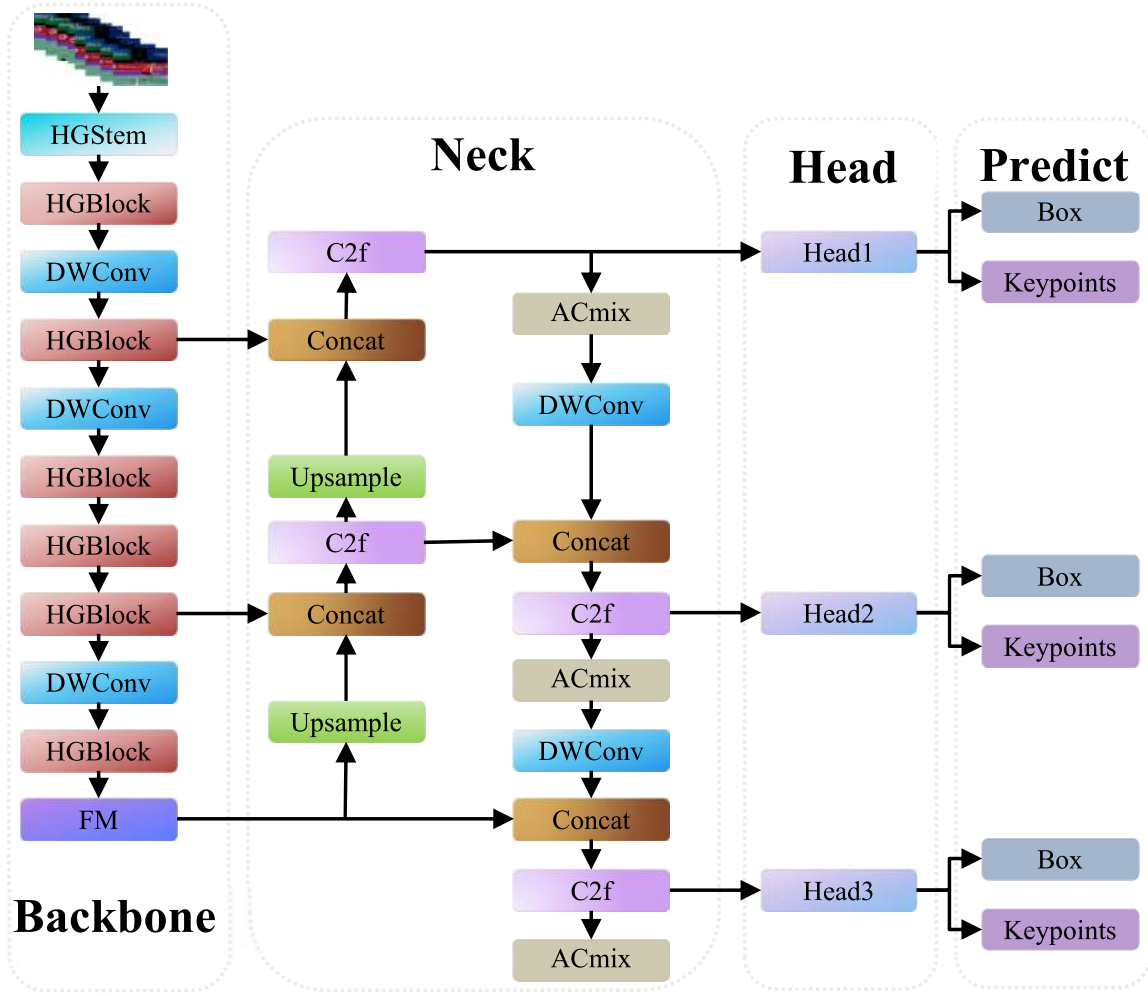


Fig. 2. Detailed architecture of YOLO-HGNet. FM stands for FocalModulation modules.

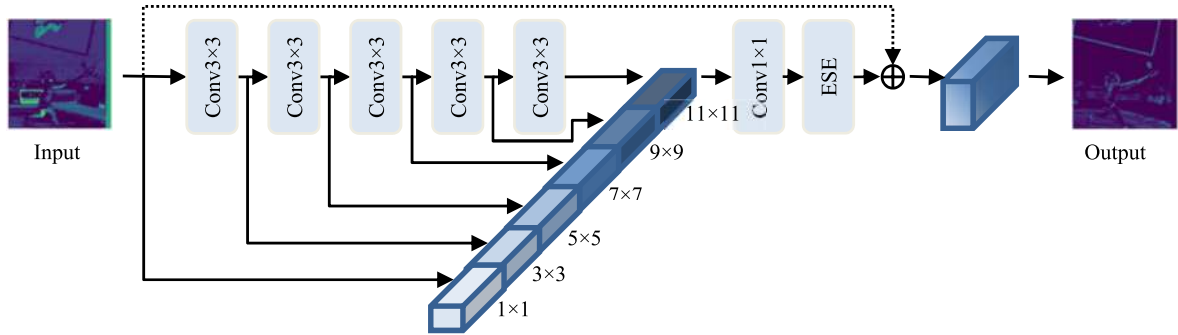


Fig. 3. Structure diagram of HGBlock(Lv et al., 2023).

applications, it is inherently inefficient for tasks where computational resources are at a premium.

Conversely, DWConv employs a more focused approach by assigning a unique convolution kernel to each channel of the input feature map. This methodology ensures that each channel is convolved independently, maintaining the exact channel count in both the input and output feature maps. Essentially, the output for each channel is the result of performing a depth-wise convolution on the corresponding input channel.

This tailored approach significantly reduces the number of parameters and computational overhead associated with the convolution operation. By eliminating cross-channel parameter integration and

focusing solely on intra-channel feature extraction, DWConv not only slashes computational requirements but also lowers energy consumption. This is particularly beneficial for models deployed on resource-constrained devices where efficiency is paramount.

3.2.3. Feature enhancement optimization

Integrating FocalModulation(Yang et al., 2022) into the YOLOv8 framework represents a pivotal advancement in enhancing the model's capability for human pose detection, especially in dynamic and complex environments. By replacing the original Fast Feature Pyramid Pooling (SPPF) module with FocalModulation, the model gains a significant boost in processing efficiency and accuracy for recognizing and locating

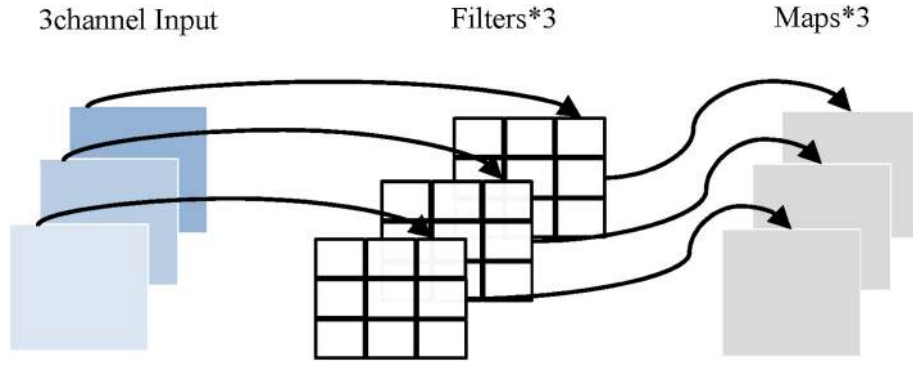


Fig. 4. The architecture of depth-wise convolution(Chollet, 2017).

human poses against intricate backgrounds.

At the heart of FocalModulation lies its dynamic focus adjustment mechanism, designed to intensify the model's perception of crucial features within an image. This innovative approach allows for a nuanced modulation of the model's focus, directing attention to critical areas and thus enabling more precise identification and localization of objects. The structure of FocalModulation, as depicted in Fig. 5, showcases its unique capacity to dynamically adapt the model's focus, ensuring that essential features are captured and emphasized effectively.

FocalModulation distinguishes itself from the SPPF module by prioritizing extracting and analyzing significant features within the image. While SPPF enhances the model's ability to handle multi-scale inputs, FocalModulation delves deeper, refining the model's focus to improve the detection of human poses and the accuracy of their localization. This targeted approach ensures that the diversity of input data is preserved, yet the emphasis is squarely on the elements most crucial for accurate pose detection.

Employing FocalModulation within the YOLOv8 architecture results

in a model that exhibits heightened sensitivity and superior performance in human pose detection tasks, particularly in scenarios involving movement. The model's enhanced adaptability to various input image sizes and its reinforced capability to pinpoint key areas significantly elevate its performance. This enhancement is invaluable across various applications, from analyzing sports events to enabling real-time surveillance, offering a robust solution for dynamic target detection and tracking in challenging settings.

3.2.4. Feature fusion optimization

Integrating the ACmix attention(Pan et al., 2022) mechanism within the neck network of the model stands as a strategic enhancement aimed at refining the feature processing capabilities. Its structure is shown in Fig. 6. This enhancement directly translates to notable improvements in the model's detection accuracy, especially when confronted with the intricacies of complex backgrounds and the challenges presented by images of varying scales. The ACmix mechanism's incorporation into the model's architecture leverages a sophisticated blend of Adaptive

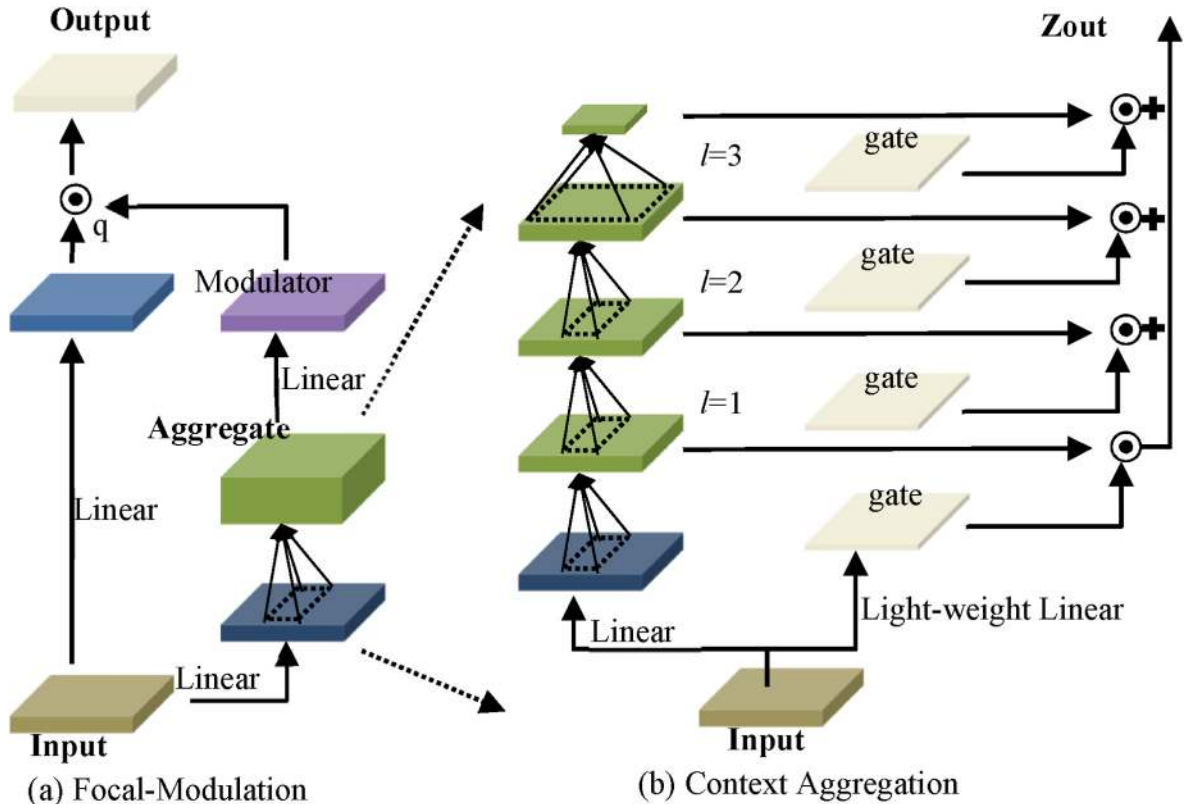


Fig. 5. The architecture of FocalModulation(Yang et al., 2022).

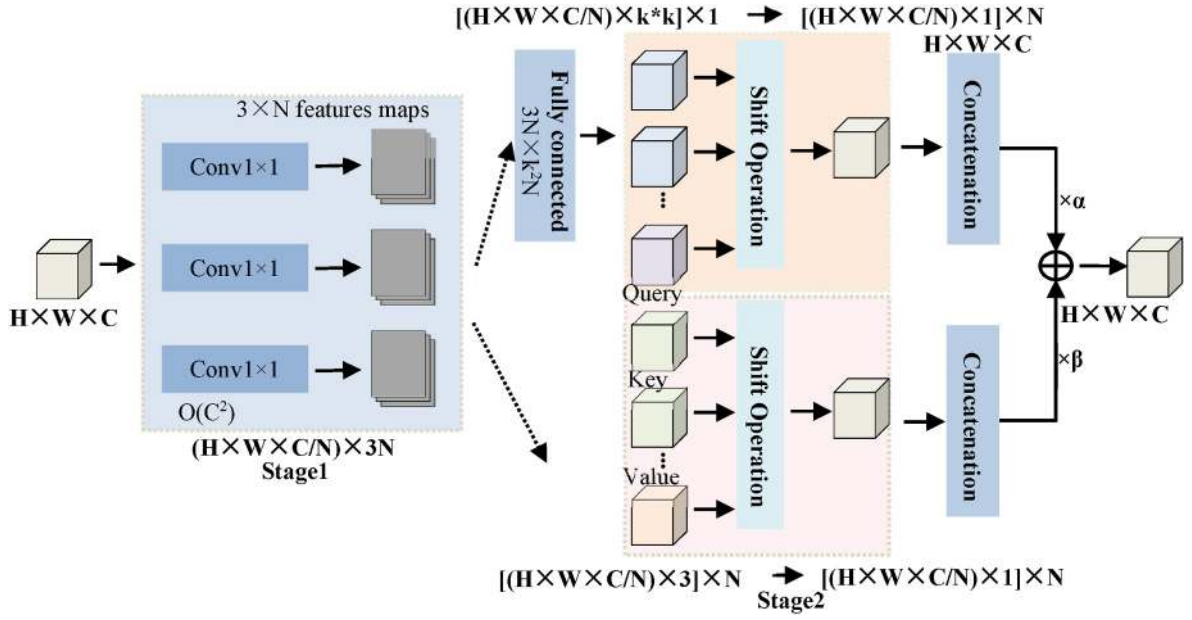


Fig. 6. ACmix(Pan et al., 2022) Hybrid Self-Attention Structure. ACmix is a collection of two powerful network advantages of convolutional networks and transformers, which have lower computational overhead while also improving network performance.

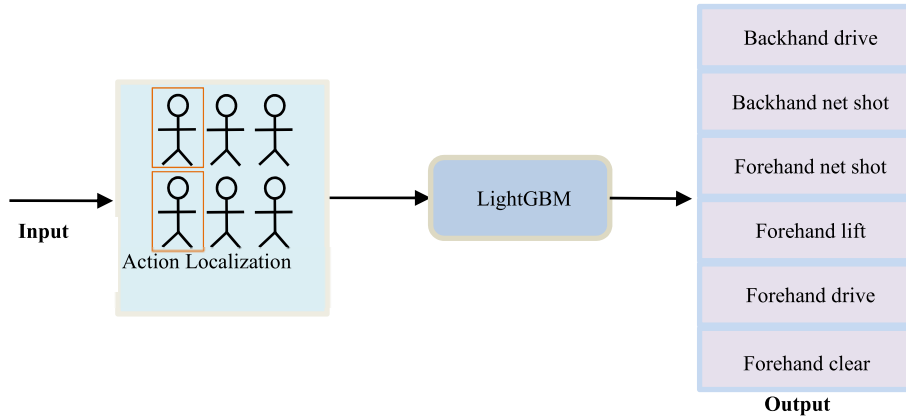


Fig. 7. Motion vectors and machine learning technique. The actions are further located by motion vectors, and then the LightGBM machine learning method is used to classify the actions.

Convolution (AC) and Mixed Attention, thus endowing the model with a heightened ability to recognize and utilize critical information within an image efficiently (see Fig. 7).

The ACmix framework emerges as a seminal innovation in deep learning, epitomizing the synthesis of convolutional neural networks (CNNs) and transformer models. This hybrid architecture capitalizes on the intrinsic strengths of both paradigms—leveraging the robust spatial feature extraction capabilities of CNNs and the dynamic sequence modeling proficiency inherent in transformers. ACmix achieves this integration with a notably reduced computational footprint, thereby not only preserving but also enhancing the overall efficiency and performance of the network.

Particularly relevant to tasks demanding a dual capacity for spatial recognition and sequence analysis, the ACmix architecture offers a compelling solution. Its ability to efficiently process intricate data sets it apart as an invaluable tool in various applications, ranging from image and video analysis to natural language processing and beyond. The ACmix model, therefore, stands as a testament to the potential of hybrid approaches in advancing the frontiers of artificial intelligence, providing a robust platform for tackling the multifaceted challenges inherent in

understanding and interpreting complex data structures.

3.2.5. Feature fusion optimization

To achieve more accurate badminton striking action localization and recognition and to optimize the model's learning efficiency and accuracy in detail, the model introduces a Keypoint Factor and Euclidean Distance(Lee et al., 2020) to establish a new loss function.

The Keypoint Factor k_f is introduced as a dynamic weighting factor. It optimizes the loss value based on the importance of keypoints in specific movements or actions, ensuring that the model pays more attention to the motion characteristics of joints during the learning process. The Euclidean distance e is introduced to quantify the difference between the model-predicted keypoint positions and their actual locations. By Calculating the straight-line distance between predicted and actual points provides an intuitive and effective metric for assessing the model's accuracy in keypoint localization. The k_f can be calculated as Equation (1):

$$k_f = \frac{\sum (k_m \neq 0) + \sum (k_m = 0)}{\sum (k_m \neq 0) + e^{-9}} \quad (1)$$

In this equation, k_m represents a keypoint mask, it can be calculated as Equation (2):

$$k_m = \frac{d}{2 \cdot (2 \cdot \sigma)^2 \cdot (A + e^{-9})} \quad (2)$$

In this equation, σ represents the standard deviation, A represents the area size, and d represents the squared Euclidean distance used to quantify the difference between the model-predicted keypoint positions and their actual locations, the formula for d would be as follows:

$$d = (p_k[\dots, 0] - g_k[\dots, 0])^2 + (p_k[\dots, 1] - g_k[\dots, 1])^2 \quad (3)$$

In this equation, p_k represents the coordinates of the predicted joint points, g_k represents the coordinates of the real joint points.

3.3. Motion vector and machine learning

To more accurately describe and quantify the direction and magnitude of movement for different parts of a player's body in space, and to grasp the overall movement of the player, this paper introduces the concept of motion vectors. Considering the symmetry of the human body, the motion vectors for the main body, upper arms, forearms, thighs, and calves on both the left and right sides of the body in the n frame are denoted as ε_{ln} , α_{ln} , β_{ln} , γ_{ln} , δ_{ln} and ε_{rn} , α_{rn} , β_{rn} , γ_{rn} , δ_{rn} . The corresponding vector coordinates are specified for each of these motion vectors.

$$\varepsilon_{ln} = h_n - lh_n = [x_n^h - x_n^{lh}, y_n^h - y_n^{lh}] \quad (4)$$

$$\alpha_{ln} = ls_n - le_n = [x_n^{ls} - x_n^{le}, y_n^{ls} - y_n^{le}] \quad (5)$$

$$\beta_{ln} = le_n - lw_n = [x_n^{le} - x_n^{lw}, y_n^{le} - y_n^{lw}] \quad (6)$$

$$\gamma_{ln} = lh_n - lk_n = [x_n^{lh} - x_n^{lk}, y_n^{lh} - y_n^{lk}] \quad (7)$$

$$\delta_{ln} = lk_n - la_n = [x_n^{lk} - x_n^{la}, y_n^{lk} - y_n^{la}] \quad (8)$$

$$\varepsilon_{rn} = h_n - rh_n = [x_n^h - x_n^{rh}, y_n^h - y_n^{rh}] \quad (9)$$

$$\alpha_{rn} = rs_n - re_n = [x_n^{rs} - x_n^{re}, y_n^{rs} - y_n^{re}] \quad (10)$$

$$\beta_{rn} = re_n - rw_n = [x_n^{re} - x_n^{rw}, y_n^{re} - y_n^{rw}] \quad (11)$$

$$\gamma_{rn} = rh_n - rk_n = [x_n^{rh} - x_n^{rk}, y_n^{rh} - y_n^{rk}] \quad (12)$$

$$\delta_{rn} = rk_n - ra_n = [x_n^{rk} - x_n^{ra}, y_n^{rk} - y_n^{ra}] \quad (13)$$

This methodology allows for a detailed analysis of the player's movements, providing insights into the efficiency, effectiveness, and potential areas for improvement in their performance. By mapping these vectors onto a coordinate system, it's possible to visualize and measure the precise movements and adjustments the player makes, offering a comprehensive tool for coaching and self-improvement in sports like badminton.

Light Gradient Boosting Machine (LightGBM)(Ke et al., 2017) is a popular gradient boosting framework known for its capability to handle large-scale data and efficient execution speed. The principle of the LightGBM model is based on the gradient boosting algorithm. It designates the target outcome labels as Y , the prediction results of the gradient boosting model, which is an ensemble of t decision trees, as $\hat{Y}^t$, and each newly grown decision tree model as f_i . The objective function of the gradient boosting algorithm can be expressed as:

$$O = L(Y, \hat{Y}^t) + \varphi(F_t) = \sum_{i=1}^n l(y_i, \hat{y}_i^t) + \sum_{i=1}^t \varphi(f_i) \quad (14)$$

In this equation, L represents the loss function, n represents the number of samples, φ represents the regularization term.

Leveraging the efficient data processing capabilities of LightGBM, this paper rapidly analyzes complex temporal sequences and spatial patterns in badminton play. By optimizing the objective function of the gradient boosting framework, each incremental decision tree model is fine-tuned to maximize the overall performance of the model, thereby enhancing the accuracy of stroke action recognition. Furthermore, the lightweight nature and optimized algorithms of LightGBM enable the model to maintain high efficiency even in resource-constrained environments. This approach capitalizes on LightGBM's strengths in handling large-scale data and computational efficiency and aligns with the goal of achieving high precision in sports analytics, particularly in the nuanced detection and analysis of badminton strokes.

3.4. The model parameters of YOLO-HGNet

To provide a clearer demonstration of the model's architecture, the parameters of the backbone and head layers of the YOLO-HGNet model are presented in Tables 1 and 2. As shown in Table 1, the backbone network combines HGBlocks and DWConv modules, which are carefully designed to ensure robust feature extraction and processing. This integration enables the model to capture intricate patterns and subtle variations in the input data, which is crucial for accurate detection and classification tasks. The HGBlocks contribute to the hierarchical feature representation, while the DWConv modules enhance the efficiency and effectiveness of the convolution operations, thereby reducing computational complexity without compromising performance.

Based on this strong foundation, as delineated in Table 2, the head layer of the YOLO-HGNet model incorporates the ACmix module. These modules are meticulously engineered to enhance the model's focus on salient features, ensuring that the most pertinent information is prioritized during the prediction phase. By integrating the attention mechanism with convolution operations, the ACmix module significantly improves the model's capacity to discern subtle distinctions in the data, resulting in more accurate and reliable final predictions. This symbiotic combination of attention and convolution enables the model to adeptly handle varying contexts and background complexities, a capability that is indispensable for real-world applications.

The synergy between the backbone layer and the head layer of the YOLO-HGNet model culminates in a robust and efficient architecture, adept at managing complex detection and classification tasks with high precision. The comprehensive parameters detailed in Tables 1 and 2 underscore the model's sophisticated design and the thoughtful engineering choices that underpin its construction. This meticulous architecture ensures that the YOLO-HGNet model not only excels in standard benchmarks but also maintains robustness and reliability across diverse and challenging environments.

Table 1
YOLO-HGNet's backbone layer parameters.

Layer	Type	Parameters
1	HGStem	[-1, 1, HGStem, [32, 48]]
2	HGBlock	[-1, 6, HGBlock, [48, 128, 3]]
3	DWConv	[-1, 1, DWConv, [128, 3, 2, 1, False]]
4	HGBlock	[-1, 6, HGBlock, [96, 512, 3]]
5	DWConv	[-1, 1, DWConv, [512, 3, 2, 1, False]]
6	HGBlock	[-1, 6, HGBlock, [192, 1024, 5, True, False]]
7	HGBlock	[-1, 6, HGBlock, [192, 1024, 5, True, True]]
8	HGBlock	[-1, 6, HGBlock, [192, 1024, 5, True, True]]
9	DWConv	[-1, 1, DWConv, [1024, 3, 2, 1, False]]
10	HGBlock	[-1, 6, HGBlock, [384, 2048, 5, True, False]]
11	FocalModulation	[-1, 1, FocalModulation, [2048]]

Note: This table reports the network structure of the HGNet backbone network, consisting of one HGStem module, one FocalModulation module, multiple HGBlocks, and multiple DWConv.

Table 2
YOLO-HGNet's head layer parameters.

Layer	Type	Parameters
1	nn.Upsample	[-1, 1, nn.Upsample, [None, 2, 'nearest']]
2	Concat	[[[-1, 7], 1, Concat, [1]]]
3	C2f	[-1, 3, C2f, [512]]
4	nn.Upsample	[-1, 1, nn.Upsample, [None, 2, 'nearest']]
5	Concat	[[[-1, 3], 1, Concat, [1]]]
6	C2f	[-1, 3, C2f, [256]]
7	ACmix	[-1, 1, ACmix, []]
8	DWConv	[-1, 1, Conv, [256, 3, 2]]
9	Concat	[[[-1, 13], 1, Concat, [1]]]
10	C2f	[-1, 3, C2f, [512]]
11	ACmix	[-1, 1, ACmix, []]
12	DWConv	[-1, 1, Conv, [512, 3, 2]]
13	Concat	[[[-1, 10], 1, Concat, [1]]]
14	C2f	[-1, 3, C2f, [1024]]
15	ACmix	[-1, 1, ACmix, []]

Note: This table reports the detection head network structure of the model, consisting of multiple Upsample, multiple Concat, multiple C2f, multiple DWConv, and multiple ACmix modules.

Furthermore, the integration of advanced modules such as HGBlocks, DWConv, and ACmix epitomizes the innovative design philosophy of the YOLO-HGNet model. Each component is meticulously optimized to enhance overall performance, ensuring that the model can effectively leverage the unique strengths of each module.

4. Materials and methods

4.1. Experimental environment and data

The experimental infrastructure for this study was established on a computing system identified as DESKTOP-JQMBK5I, which is powered by an AMD EPYC 7643 48-Core Processor operating at a frequency of 2.30 GHz, augmented by four processors and equipped with 32.0 GB of Random Access Memory (RAM). This setup, running on a Windows operating system, provides a robust computational platform designed to meet the rigorous demands associated with training complex neural network models on extensive datasets. The experiments leveraged Python 3.8 alongside the PyTorch deep learning library, version 1.13.1, for model development and training processes. Model optimization was facilitated through the Adam optimizer, with an initial learning rate established at 0.00015. To enhance training efficacy, the learning rate was programmed to decrement by a factor of 10 subsequent to every 20 epochs, culminating in a comprehensive training duration of 500 epochs to achieve model convergence.

In parallel to configuring an optimal experimental setup, rigorous steps were undertaken to ensure the integrity, relevance, and computational manageability of the dataset, as shown in Fig. 8, which was

amalgamated from internet sources and the publicly accessible "badminton_stroke_video" dataset on Kaggle. As shown in Fig. 9, This dataset encompasses six distinct categories of badminton strokes, featuring 100–120 videos per category and averaging 60 frames per video. To refine the dataset's quality, enhance computational efficiency, and mitigate information redundancy, the following preparatory procedures were executed.

- **Key frame extraction:** Selective extraction of key frames from video files constituted a critical component of the dataset preparation, aimed at capturing pivotal moments of action for analysis.
- **Background diversity enhancement:** In order to enhance the background variability of the dataset, images with complex backgrounds are selected from various Internet sources, thus enriching the contextual spectrum of the dataset.
- **Data Augmentation:** The application of data augmentation techniques was employed to bolster the dataset's diversity and robustness, preparing the model to generalize across varied scenarios.
- **Manual Annotation:** Detailed manual annotation of the collated images was meticulously performed to ensure the highest level of accuracy. This process involved delineating 13 key pose nodes for each player, along with detection boxes that accurately represent the players' striking actions within the dataset.

By providing precise and comprehensive annotations, this dataset serves as a robust foundation for training and evaluating models aimed at understanding and predicting player actions in badminton.

4.2. Results on test set

Table 3 presents a comprehensive comparative analysis of the proposed model against established methods, including those based on YOLOv5, Lightopenpose, YOLOv7 and so on. This comparison primarily focuses on the metric of mean Average Precision (mAP), a standard measure used to evaluate the accuracy of object detectors. The results highlight the competitive performance of the proposed model across various benchmarks. Notably, the model demonstrates impressive achievements in mAP at an Intersection over Union (IoU) threshold of 50% (mAP50), indicating its ability to accurately detect and classify objects within the given dataset. The comparative analysis underscores the advancements made by the proposed model in addressing the challenges posed by previous methods, offering a significant improvement in detection accuracy and reliability.

Moreover, the results in Table 3 demonstrate the model's ability to outperform established methods consistently. The enhancements incorporated into the model not only improve its precision and recall but also enable it to operate efficiently in real-world scenarios. This makes the proposed model a valuable tool for various applications that require



Fig. 8. Image dataset, which consists of images with complex backgrounds collected from the Internet, Google, BWF, and other public websites.



Fig. 9. The publicly available dataset on Kaggle consists of six types of action videos, a,b,c,d,e,f denote backhand drive, backhand drive, forehand clear, forehand drive, forehand lift, and forehand net, respectively Sequences of shot Actions The public dataset on Kaggle consists of six types of action videos, a,b,c,d,e,f denote the sequences of actions backhand drive, backhand drive, forehand clear, forehand drive, forehand lift, and forehand net shot, respectively.

accurate and reliable object detection.

The superior performance of the proposed model, as evidenced by its mean Average Precision (mAP) values, can be primarily attributed to the integration of three pivotal advancements: a novel attention mechanism, an innovative feature pyramid pooling model, and an innovative backbone network. These enhancements collectively contribute to the

model's heightened ability to accurately discern and analyze the dynamic regions of interest within the human body, facilitating a more nuanced understanding of the body's structure and the intricate relationships between its components.

Table 3

Compared with the existing methods on the test data set.

Method	Precision	Recall	mAP50	mAP50-95
YOLOv5pose(Huang et al., 2021)	83.5	79.6	80.2	44.9
Lightopenpose(Osokin, 2018)	76.5	77.6	72.5	37.4
HigherHRNet(Cheng et al., 2020)	72.3	71.3	85.6	36.2
Openpose(Cao et al., 2017)	74.6	65.9	69.3	39.5
YOLO-Rlepose(Jiang et al., 2024)	84.0	81.2	78.4	47.4
YOLOv7-Pose(Monnikhof et al., 2023)	86.9	85.4	84.3	53.8
YOLOv8n-Pose(Wang et al., 2023)	89.9	87.5	85.0	47.2
YOLOv8s-Pose(Wang et al., 2023)	90.7	88.4	87.3	55.3
YOLOv8m-Pose(Wang et al., 2023)	91.1	89.6	87.2	58.0
HRformer(Yuan et al., 2021)	78.9	86.1	90.7	79.6
RTMPose(Jiang et al., 2023)	68.6	75.5	71.3	44.2
ED-pose(Yang et al., 2023)	87.6	81.6	91.5	73.3
Ours	95.7	95.8	96.1	69.5

Note: This table reports the recognition performance of different models in badminton hitting action. Four common indicators in the field Precision, Recall, mAP50 and MAP50-95 are used as evaluation indicators.

4.3. Comparison experiments of different classifiers

To substantiate the efficacy of the machine learning methodologies employed as classifiers in this study, a series of comparative experiments were conducted utilizing the "badminton_stroke_video" dataset. These experiments were designed to benchmark the performance of the proposed method against two widely recognized machine learning techniques: XGBoost and CatBoost. The results of these comparative analyses are systematically presented in Table 4.

The findings elucidate that the classifier implemented in this research outperforms the benchmarked methods, achieving an exemplary average accuracy of 95.4%. This superior performance underscores the method's proficiency in the precise classification of badminton striking actions, thereby affirming its effectiveness as a tool for action recognition within the domain of sports analytics.

Such a high degree of accuracy highlights the method's robustness and its potential applicability in real-world scenarios where the accurate differentiation of complex actions is paramount. The comparative analysis validates the superiority of the selected machine learning classifiers in this paper and demonstrates their practical utility in enhancing the analytical precision of badminton action recognition.

The success of the proposed method can be attributed to several factors. First, the meticulous feature engineering process ensured that the most relevant aspects of the badminton strokes were captured and used in the classification process. Second, the integration of advanced machine learning techniques allowed for the effective handling of the intricate patterns and variances within the dataset. Finally, the rigorous training and validation processes employed ensured that the model was well-tuned and capable of generalizing to new, unseen data.

4.4. Badminton hitting action recognition experiment

To rigorously assess the recognition capabilities of the YOLO-HGNet network model, an enhanced derivative of YOLOv8, this study conducts

Table 4

In the comparison table with different classifiers, the classifier selected in this paper has the best effect.

Method	Recall	Precision	F1_Score
XGBoost	93.4%	94.2%	93.4
CatBoost	89.3%	88.9%	89.6
Ours	94.7%	95.4%	95.1

Note: This table reports the performance of different mainstream machine learning methods for badminton action classification. The evaluation index is composed of three main indicators in the field Recall, Precision and F1_Score.

a series of experiments focusing on badminton striking actions using the "badminton_stroke_video" dataset. For a more detailed analysis, the dataset is meticulously categorized into six distinct types of badminton strokes: backhand push ball, backhand net front small ball, forehand high ball, forehand push ball, forehand pick ball, and forehand net front small ball. The outcomes of these recognition experiments are systematically compiled in Table 5.

The experimental results reveal that the model demonstrates exceptional precision and recall rates, exceeding 90% across all categories of badminton strokes. This exceptional performance underscores the model's adeptness at navigating complex scenarios and recognizing a diverse array of actions with high accuracy. Notably, the model achieves an average recall rate of 94.7% and an average precision of 95.4%, signifying its remarkable efficiency in accurately identifying and differentiating between various badminton striking actions.

Such high levels of recall and precision indicate the proposed model's potential to serve as a valuable analytical tool for coaches and sports science researchers. By providing precise action recognition capabilities, the model can significantly contribute to optimizing training regimes and enhancing athletes' technical proficiencies. Furthermore, the insights garnered from the model's analysis have the potential to elevate competitive performance and mitigate the risk of sports-related injuries, offering a multifaceted benefit to the domain of sports analytics.

Fig. 10 displays the confusion matrix for the recognition of badminton striking actions, providing a detailed visualization of the model's performance across different action categories. From an analysis of the matrix, it is evident that the model achieves a high degree of accuracy, with the correctly identified video samples for each category constituting the majority of observations. Out of a total of 838 observed videos, the model accurately predicted 800, resulting in an impressive overall recognition accuracy of 93.8%. This high accuracy rate underscores the model's effectiveness in badminton action recognition tasks.

Despite the model's overall success, a closer examination of the confusion matrix reveals certain areas of misclassification, particularly between actions that bear a semblance to one another. Notably, a higher incidence of confusion was observed between the backhand push and the backhand net front small ball, as well as between the forehand lob and the forehand push. This misclassification is primarily attributed to the occasional overlap in the visual characteristics of these actions. Specifically, the similarity in the execution of the backhand push and the backhand net front small ball, along with the resemblance between the forehand lob and the forehand push, poses challenges to the model's differentiation capabilities.

The identification of these confusion points is critical for understanding the limitations of the current model and serves as a foundation for future improvements. Addressing these specific areas of confusion, especially in actions with subtle differences, remains a pivotal area for refinement to enhance the model's overall performance in action recognition.

Table 5

Six different batting action recognition data sheet. The method used in this paper achieves an average accuracy of 95.4% for the recognition of six batting actions.

Method	Recall	Precision	F1_Score
backhand drive	93.7%	91.0%	92.4
backhand net shot	90.0%	95.8%	92.6
forehand clear	93.8%	97.5%	94.3
forehand drive	96.9%	95.0%	95.5
forehand lift	97.8%	96.0%	96.9
forehand net shot	96.2%	97.2%	96.8
Average	94.7%	95.4%	94.8

Note: This table reports the classification results of the proposed model for six different batting actions. The evaluation index is composed of three main indicators.

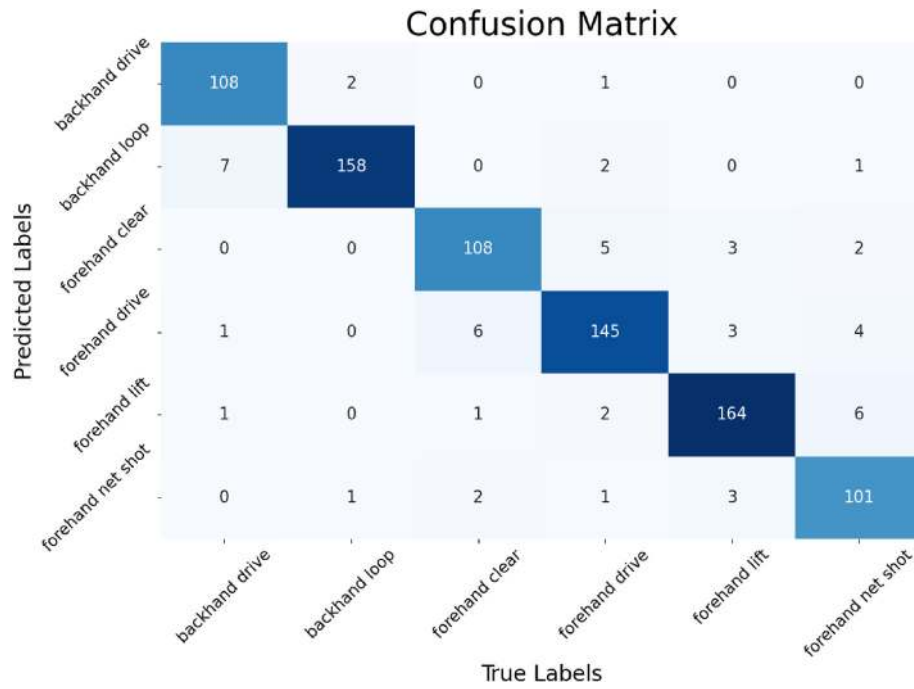


Fig. 10. Six types of batting action recognition confusion matrix. The method used in this paper has a significant recognition effect and can basically correctly identify the badminton hitting action.

To mitigate these misclassifications, future work could focus on several key areas. Firstly, enhancing the feature extraction process to better capture the nuanced differences between similar actions could improve the model's accuracy. Secondly, incorporating additional contextual information, such as the player's position on the court and the trajectory of the shuttlecock, could provide further differentiation cues. Lastly, employing advanced techniques such as fine-grained action recognition or multi-modal learning, which integrates visual and non-visual data, could also contribute to reducing confusion between closely related actions.

4.5. Ablation experiments

To ascertain the effectiveness of the YOLO-HGNet badminton hitting action recognition network model, predicated on the enhancements to YOLOv8 delineated in this manuscript, a series of ablation experiments were executed on our dataset. These experiments were designed to

systematically evaluate the impact of each proposed improvement on the model's performance. The specific configurations tested are in Table 6.

Ablation experiments are crucial for understanding the contribution of each individual component to the overall performance of the model. By selectively removing or modifying certain aspects of the model, we can isolate their effects and determine their relative importance. In this study, we focused on evaluating the following enhancements: the novel attention mechanism, the innovative feature pyramid pooling model, and the advanced backbone network.

The results of these experiments, detailed in Table 6, provide a clear indication of the effectiveness of each enhancement. For instance, the inclusion of the novel attention mechanism significantly improved the model's ability to focus on critical features, leading to higher precision and recall rates. Similarly, the feature pyramid pooling model contributed to better multi-scale feature extraction, which is essential for accurately recognizing actions of varying sizes and motions. Lastly, the

Table 6

Ablation experiments were carried out on the self-built dataset.

Backbone	ACmix	FocalModulation	SPPF	Conv	DWConv	mAP50
CSPDarkNet	×	×	✓	✓	×	0.858
	✓	×	✓	✓	×	0.876
	×	✓	×	✓	×	0.813
	✓	✓	×	✓	×	0.881
	×	×	✓	×	✓	0.862
	✓	×	✓	×	✓	0.886
	×	✓	×	×	✓	0.898
	✓	✓	×	×	✓	0.826
HGNet	×	×	✓	✓	×	0.875
	✓	×	✓	✓	×	0.896
	×	✓	×	✓	×	0.901
	✓	✓	×	✓	×	0.913
	×	×	✓	×	✓	0.922
	✓	×	✓	×	✓	0.935
	×	✓	×	×	✓	0.926
	✓	✓	×	×	✓	0.961

Note: This table the performance of the model to detect i for the batting action when different model structures are introduced, where ✓ means that the model structure is cited and × means that the model structure is forbidden.

advanced backbone network provided a robust foundation for feature extraction, enhancing the model's overall efficiency and accuracy.

Through these ablation studies, we were able to demonstrate that each proposed enhancement contributes positively to the model's performance. The cumulative effect of these improvements is reflected in the model's superior recognition capabilities, as evidenced by the high precision and recall rates reported in the experimental results.

Fig. 11, as detailed in Table 6, delineates the outcomes of the ablation experiments conducted to evaluate the performance of the proposed YOLO-HGNet model against the baseline YOLOv8 model. In this comprehensive analysis, the YOLO-HGNet model, embodying the combined enhancements of the ACmix and ASPP modules integrated into the YOLOv8 framework, demonstrates a notable performance increase of 10.3% over the original YOLOv8 model.

This significant enhancement is quantifiable in terms of performance metrics and manifestly evident in the visual results derived from the model's application. A key aspect of this improvement is observed in the precision of keypoint prediction, where the YOLO-HGNet model exhibits superior accuracy. This increase in precision underscores the effectiveness of the modifications introduced to the YOLOv8 model, affirming the value of the backbone, the ACmix self-attention mechanism, and the FocalModulation module in refining the model's capacity for accurate action recognition.

The results, as shown in Fig. 11, highlight the tangible benefits of these enhancements. The YOLO-HGNet model not only outperforms the baseline YOLOv8 model but also demonstrates a higher level of robustness and reliability in real-world applications. This improvement is particularly important for sports analytics, where precise action recognition is essential for optimizing training regimes, enhancing athlete performance, and preventing injuries.

5. Conclusions

In this paper, we integrate advanced technologies such as HGNet, the ACmix attention mechanism, and FocalModulation into the YOLOv8 model to develop a new model named YOLO-HGNet, which represents a significant advancement in athlete pose detection technology. We employ HGNet as the backbone network to enhance the detection efficiency and accuracy of the entire system, while the integration of the ACmix attention mechanism into feature-rich nodes improves the model's data processing capabilities. Furthermore, the incorporation of the new feature processing method, FocalModulation, into the model's feature processing node enhances its feature extraction capabilities.

Our method was compared with several current mainstream approaches, including Openpose, YOLO-Rlepose, and YOLOv7-Pose. Experimental results demonstrate that our model achieves the highest Mean Average Precision (mAP) in the pose detection of badminton players, indicating superior performance in badminton action detection. Additionally, we utilize action vectors and machine learning methods to classify six types of actions, achieving an impressive accuracy of 95.4%, which is nearly equivalent to human discrimination.

Based on our experimental findings, we offer several recommendations for researchers in the field of badminton action research. Firstly, the selection of backbone networks for YOLO models is critical, as it enhances the model's feature extraction capabilities and aids in identifying key features. Secondly, object detection algorithms applied to action recognition exhibit excellent performance and can be leveraged to assist in player training.

Future research will focus on several key areas to enhance the YOLO-HGNet model and broaden its applications. We will prioritize refining the model's architecture to reduce parameter count and improve real-time processing efficiency, employing techniques such as pruning, quantization, and knowledge distillation. Additionally, optimizing

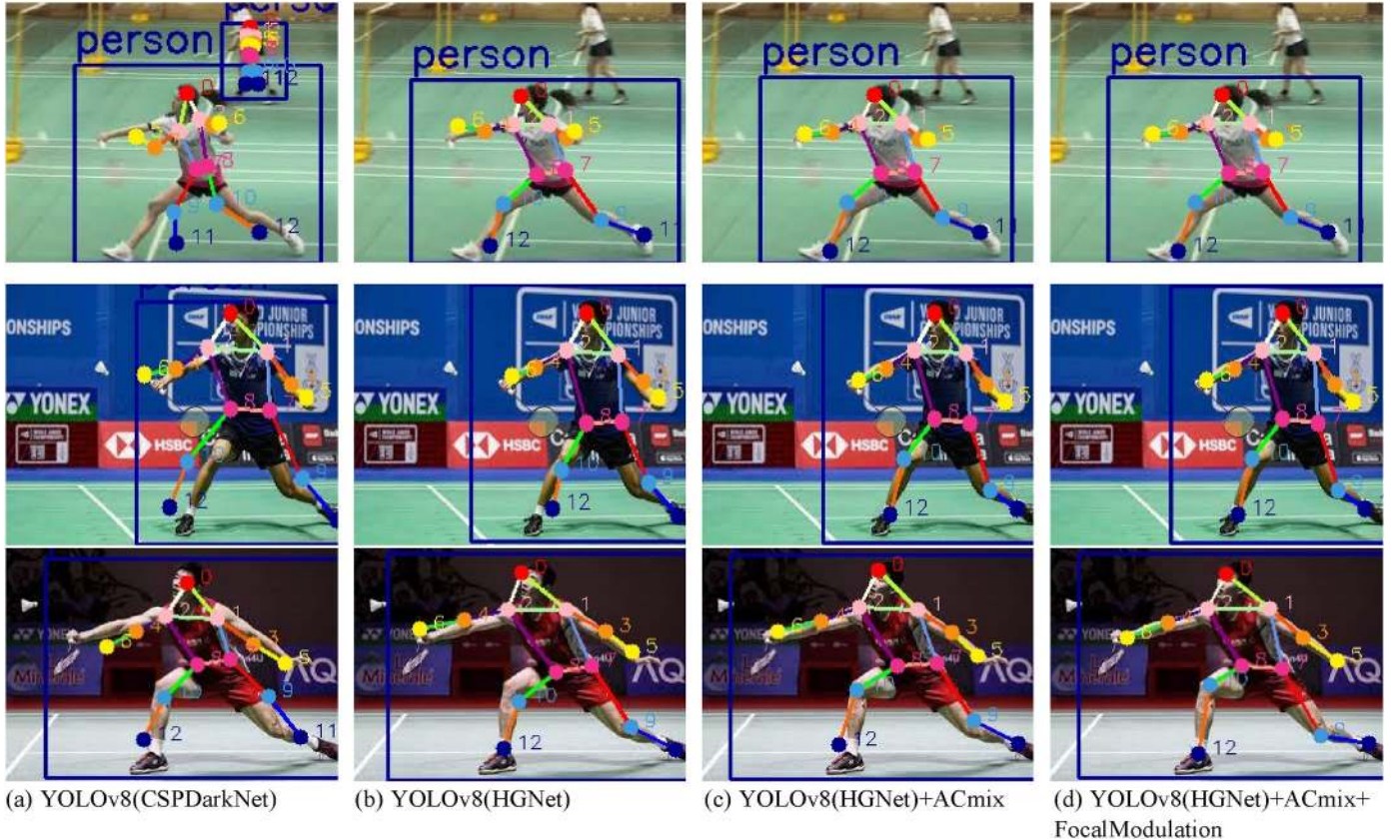


Fig. 11. Comparison of results from ablation experiments.

YOLO-HGNet for deployment on edge computing devices is essential, involving adaptations for various hardware platforms and operating systems to ensure effective and responsive performance. Expanding the model's applicability beyond badminton to include other sports and physical activities will be explored, along with integrating additional data modalities like depth sensing and wearable sensors to enhance action recognition accuracy. We also plan to incorporate augmented data collection methods and develop new evaluation metrics to assess model robustness and performance comprehensively. Furthermore, practical applications will be investigated, including the development of training assistance tools and collaboration with sports science experts to validate and refine the model based on real-world needs and feedback. These efforts aim to advance YOLO-HGNet's capabilities and contribute significantly to the fields of action recognition and sports science.

CRediT authorship contribution statement

Wenwen Yang: Writing – original draft, Visualization, Software, Resources, Methodology, Data curation. **Minlan Jiang:** Writing – review & editing, Supervision, Project administration, Investigation, Formal analysis, Conceptualization. **Xiaosheng Fang:** Writing – review & editing, Investigation, Formal analysis, Conceptualization. **Xiaowei Shi:** Resources, Investigation, Formal analysis, Conceptualization. **Yizheng Guo:** Visualization, Validation, Investigation, Data curation. **Mohammed A.A. Al-qaness:** Writing – review & editing, Validation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

We used public datasets as described in the paper.

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References

- Bochkovskiy, A., Wang, C.-Y., Liao, H.-Y.M., 2020. Yolov4: Optimal Speed and Accuracy of Object Detection. *arXiv preprint arXiv:2004.10934*.
- Cao, Z., Simon, T., Wei, S.-E., Sheikh, Y., 2017. Realtime multi-person 2d pose estimation using part affinity fields. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 7291–7299.
- Chen, Y.-H., 2023. A new perspective for shuttlecock hitting event detection. *arXiv preprint arXiv:2306.10293*.
- Chen, Y., 2024. Action detection in badminton courts using AVA dataset and MMAAction2 architecture with slow Fast model. *Highlights in Science, Engineering and Technology* 85, 783–789.
- Cheng, B., Xiao, B., Wang, J., Shi, H., Huang, T.S., Zhang, L., 2020. Higherhrnet: scale-aware representation learning for bottom-up human pose estimation. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 5386–5395.
- Chollet, F., 2017. Xception: deep learning with depthwise separable convolutions. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1251–1258.
- Chu, W.-T., Situmeang, S., 2017. Badminton video analysis based on spatiotemporal and stroke features. In: *Proceedings of the 2017 ACM on International Conference on Multimedia Retrieval*, pp. 448–451.
- Feichtenhofer, C., Pinz, A., Wildes, R.P., 2017. Spatiotemporal multiplier networks for video action recognition. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 4768–4777.
- Goh, G.L., Goh, G.D., Pan, J.W., Teng, P.S.P., Kong, P.W., 2023. Automated Service Height Fault Detection Using Computer Vision and Machine Learning for Badminton Matches 23, 9759.
- He, Z., Liu, G., Zhang, B., Ye, B., Zhu, H., 2024. Impact of specialized fatigue and backhand smash on the ankle biomechanics of female badminton players. *Sci. Rep.* 14, 10282.
- Host, K., Ivašić-Kos, M., 2022. An overview of human action recognition in sports based on computer vision. *Heliyon* 8, e09633.
- Huang, T., Li, Y., Zhu, W., 2021. An auxiliary training method for single-player badminton. In: *2021 16th International Conference on Computer Science & Education (ICCSE)*, pp. 441–446.
- Jiang, T., Lu, P., Zhang, L., Ma, N., Han, R., Lyu, C., Li, Y., Chen, K., 2023. RtmPose: real-time multi-person pose estimation based on mmpose. *arXiv preprint arXiv:2303.07399*.
- Jiang, Y., Yang, K., Zhu, J., Qin, L., 2024. YOLO-rlepose: improved YOLO based on swin transformer and rle-oks loss for multi-person pose estimation. In: *Electronics*, vol. 13.
- Jocher, G., 2020. Ultralytics YOLOv5.
- Jocher, G., Chaurasia, A., Qiu, J., 2023. Ultralytics YOLOv8.
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu, T.-Y., 2017. Lightgbm: a highly efficient gradient boosting decision tree. *Adv. Neural Inf. Process. Syst.* 30.
- Lee, B.S., Phattharaphon, R., Yean, S., Liu, J., Shakya, M., 2020. Euclidean distance based loss function for eye-gaze estimation. In: *2020 IEEE Sensors Applications Symposium (SAS)*, pp. 1–5.
- Lin, T., Aouididi, A., Chen, Z., Beyer, J., Pfister, H., Wang, J.H., 2024. VIRd: immersive match video analysis for high-performance badminton coaching. *IEEE Trans. Visual. Comput. Graph.* 30, 458–468.
- Liu, C., 2024. A predictive model: the key to success for badminton servers and receivers. *bioRxiv* 2024, 579788, 02.10.
- Liu, T., Shao, M., Yin, D., Li, Y., Yang, N., Yin, R., Leng, Y., Jin, H., Hong, H., 2017. The effect of badminton training on the ability of same-domain action anticipation for adult novices: evidence from behavior and ERPs. *Neurosci. Lett.* 660, 6–11.
- Liu, W., Ke, J., 2020. A brief analysis of multi-ball training in badminton. *Education Research Frontier* 10.
- Lv, W., Xu, S., Zhao, Y., Wang, G., Wei, J., Cui, C., Du, Y., Dang, Q., Liu, Y., 2023. Detrs beat yolos on real-time object detection. *arXiv preprint arXiv:2304.08069*.
- Magnus, L., Stella, G., Dan, W., Pascal, M., 2023. TempPose: a new skeleton-based transformer model designed for fine-grained motion recognition in badminton. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 5198–5207.
- Maji, D., Nagori, S., Mathew, M., Poddar, D., 2022. Yolo-pose: enhancing yolo for multi person pose estimation using object keypoint similarity loss. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 2637–2646.
- Monnikhof, K.O., Areerob, P., Wu, Z., Tanasnitikul, T., Kumwilaisak, W., 2023. Novel personal protective equipment detection technique with attention-based YOLOv7 and human pose estimation. *APSIPA Transactions on Signal and Information Processing* 12.
- Osokin, D., 2018. Real-time 2d multi-person pose estimation on cpu: lightweight openpose. *arXiv preprint arXiv:1811.12004*.
- Pan, X., Ge, C., Lu, R., Song, S., Chen, G., Huang, Z., Huang, G., 2022. On the Integration of Self-Attention and Convolution, pp. 815–825.
- Ramasinghe, S., Chathuramali, K.G.M., Rodrigo, R., 2014. Recognition of badminton strokes using dense trajectories. In: *7th International Conference on Information and Automation for Sustainability*, pp. 1–6.
- Redmon, J., Divvala, S., Girshick, R., Farhadi, A., 2016. You only look once: unified, real-time object detection. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 779–788.
- Redmon, J., Farhadi, A., 2017. YOLO9000: better, faster, stronger. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 7263–7271.
- Redmon, J., Farhadi, A., 2018. YOLOv3: an Incremental Improvement. *arXiv preprint arXiv:1804.02767*.
- Vijay, Singh Dr, Sameer, A., 2023. An imperial study on the importance of body alignment and its impact on badminton smash training. *International journal of economic perspectives* 17, 254–261.
- Wang, S., Zhang, X., Ma, F., Li, J., Huang, Y., 2023. Single-stage pose estimation and joint angle extraction method for moving human body. *Electronics* 12, 4644.
- Yang, J., Li, C., Dai, X., Gao, J., 2022. Focal modulation networks. *Adv. Neural Inf. Process. Syst.* 35, 4203–4217.
- Yang, J., Zeng, A., Liu, S., Li, F., Zhang, R., Zhang, L., 2023. Explicit box detection unifies end-to-end multi-person pose estimation. *arXiv preprint arXiv:2302.01593*.
- Yuan, Y., Fu, R., Huang, L., Lin, W., Zhang, C., Chen, X., Wang, J., 2021. Hrformer: high-resolution vision transformer for dense predict. *Adv. Neural Inf. Process. Syst.* 34, 7281–7293.
- Yue, S.S., Paramesran, R., Krishnasamy, G., 2024. Ready-to-Serve detection in badminton videos. In: *2024 International Conference on Electronics, Information, and Communication (ICEIC)*, pp. 1–5.
- Zhao, J., Masood, R., Seneviratne, S., 2021. A review of computer vision methods in network security. *IEEE Communications Surveys & Tutorials* 23, 1838–1878.
- Zhao, L., Zhao, Z., Zhang, E., Hawbani, A., Al-Dubai, A.Y., Tan, Z., Hussain, A., 2023. A digital twin-assisted intelligent partial offloading approach for vehicular edge computing. *IEEE J. Sel. Area. Commun.* 41, 3386–3400.
- Zheng, H., Shen, H., Zhang, Y., Wang, H., 2023. Badminton action recognition based on improved I3D convolutional neural network. *3rd International Conference on Artificial Intelligence, Automation, and High-Performance Computing (AIAHPC 2023)* 12717, 776–784.